Yunhao (Henry) Xu

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EDUCATION

Carnegie Mellon University (CMU) **Aug 2024 - May 2026**
M.S. in Materials Science | GPA: 3.9/4.0
Core Courses: Structure and Characterization of Materials, Thermodynamics of Materials, Advanced Polymer Science, 3D Printing

Shanghai Institute of Technology (SIT) **Sep 2020 - Jun 2024**
B.Eng. in Materials Science and Engineering | GPA: 3.9/5.0
Core Courses: Fundamentals of Materials Science, Polymer Physics, Polymer Chemistry, Polymer Processing Techniques

SKILLS

- **Engineering & Manufacturing:** CATIA, AutoCAD, SolidWorks, ANSYS, 3D Modeling, 3D Printing, Failure Analysis, DOE
- **Data Analysis & Programming:** Python, SQL, MATLAB, JMP, Tableau, Advanced Excel
- **Materials Characterization & Testing:** TEM, SEM, DSC, Tensile Testing, Rotational Viscometer
- **Laboratory:** Wet Lab, Polymer Synthesis

PROFESSIONAL EXPERIENCE

Fuyao Glass America Inc. [Dayton, OH] **Jul 2026 – Present**
Packaging / Process Engineer

- Led packaging development for Tesla and BMW automotive glass programs, translating product and logistics requirements into CATIA-based packaging designs and coordinating cross-functional design reviews through production release.
- Validated prototype racks and packaging systems through dimensional, fit, and functional reviews; identified dunnage, flipper, and product-loading issues and drove supplier corrective actions.
- Developed packaging specifications, inspection standards, and work instructions to standardize packaging quality and support mass production.
- Designed custom fixtures and protective components using CATIA and 3D printing to improve product protection and operator efficiency.

R&D Intern, Shanghai Chest Medical Devices Co. [Shanghai, China] **Jul 2024 - Aug 2024**

- Created 2D and 3D CAD drawings of medical devices and generated 2D drawings for inner and outer packaging, ensuring design accuracy for downstream testing and analysis.
- Performed bursting strength, hydrostatic head, and dry state fiber shedding tests on medical devices, collecting structured datasets for quality evaluation.
- Analyzed over 150 test samples using Python (Pandas, NumPy) and Excel, applying statistical methods and visualization to identify key quality trends and reducing data processing time by 40% through automation.
- Collaborated with R&D, QA, and Manufacturing teams to investigate test inconsistencies; applied root cause analysis and data-driven insights to resolve 2 major issues, reducing retesting frequency by 25% and enabling faster design iterations.

RESEARCH EXPERIENCE

ML-Based Prediction of Mechanical Properties in Polymer-Grafted Nanoparticles, CMU **Aug 2025 – Present**

- Developed 8 regression models to predict Young's modulus of polymer-grafted nanoparticles from molecular structure parameters (degree of polymerization, grafting density).
- Implemented rigorous hyperparameter optimization using leave-one-out cross-validation with MAPE as the objective function, training models on PMMA dataset (80/20 train-test split).
- Demonstrated the models' capability to learn structure-property relationships in different polymer systems (PMMA data for training and PS for validation).
- Created comprehensive analysis framework with automated scripts for model training, performance benchmarking, cross-validation, and visualization generation, producing publication-ready figures and detailed performance metrics tables.

Self-Healing Hybrid Materials & Thin-Film Characterization, CMU **Aug 2024 – Aug 2025**

- Optimized thin-film coating processes ($oSiO_2$–PMMA self-healing films via spin-coating with PAA sacrificial layers).
- Performed TEM analysis. Quantified hyperuniformity in microstructural patterns using Python, imageJ and MATLAB.
- Designed workflows to track experimental parameters and discovered the correlation between coating conditions and healing performance.
- Established collaborations with advisors and cross-disciplinary peers to refine methodologies. Manuscript in preparation.

Water-Based Acrylate Pressure-Sensitive Adhesives, SIT **Jan 2024 - Jun 2024**

- Synthesized waterborne acrylic PSAs with varied structures using suspension polymerization.
- Characterized material properties through standardized tests: solid content (oven method), tack testing (rolling ball), shear adhesion, 180° peel strength (tensile tester), and viscosity (rotational viscometer).
- Established correlations between formulations and performance metrics, accelerating identification of optimal compositions.
- Designed Tableau visualizations to present structure-property relationships, enabling design improvements.
- Optimized formulations by tuning functional monomer/initiator types, soft–hard monomer ratios, Tg, and particle structures, achieving deeper understanding of structure–property-performance relationships in PSAs.